



C. Watering the Flowerbed

CPU: 1.00s

Memory: 2048MB

In the fertile and magical lands of Numeria, a legendary garden is entrusted to your care. This garden contains N **flowerbeds**, each filled with delicate and precious flowers that require precise watering to stay healthy. However, these flowers are sensitive, and maintaining their well-being is no simple task.

Each flowerbed has two critical watering requirements:

For, i^{th} flowerbed, if it is not watered within X_i days since its last watering, the plants will dry out and die.

Also, if the i^{th} flowerbed is watered again within Y_i days since its last watering, the excess water will harm the roots and plants may die.

All the flowers were watered today. You have to take care of the garden for K more days. Find out the minimum total number of **times** you need to water to ensure all flowerbeds remain healthy after K **days**. On any day, if you water multiple flowerbeds, you have to count them separately.

Input Description

The input consists of multiple test cases.

The first line contains an integer $T(1 \leq T \leq 25)$, the number of test cases. For each test case, The first line contains two integers $N(1 \leq N \leq 1000)$ and $K(1 \leq K \leq 10^9)$, where N denotes the number of flowerbeds and K denotes the number of days you need to take care of the flowerbeds. The next N lines each contain two integers, X_i, Y_i where $(1 \leq Y_i < X_i \leq 10^9)$.

Output Description

For each test case, print the case number followed by the total number of times you need to water.

Samples

Input

Output

1
2 10
3 1
6 3

Case 1: 4

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Ends in
03:30:08

THE CONTEST IS RUNNING

SUBMIT

Language

C++

-O2 -static -std=gnu++17

SELECT SOURCE

SUBMIT

MY SUBMISSIONS



No submissions yet!